

Harnessing the power of single-cell large language models with parameter-efficient fine-tuning using scPEFT

Fei He

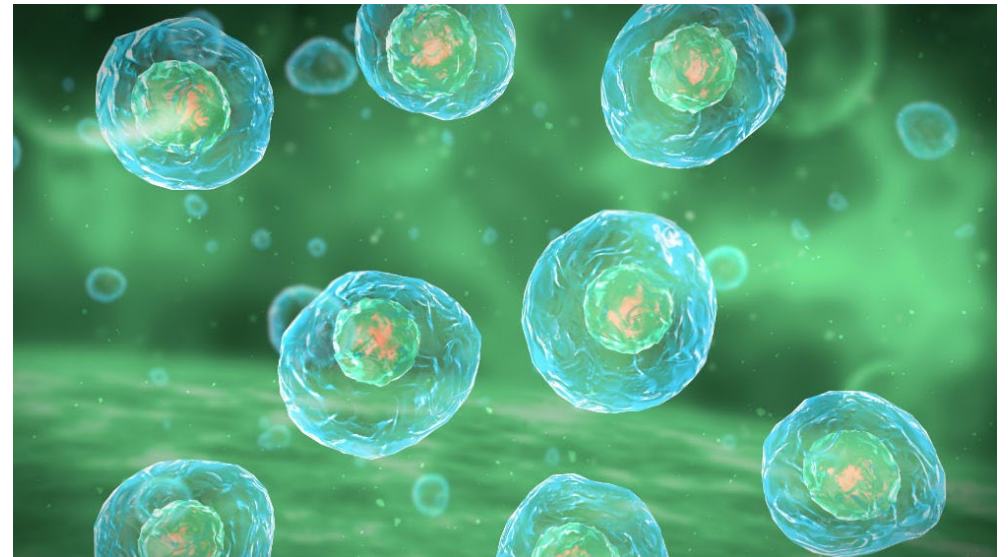
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Single cell Large Language Models (scLLMs)

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scGPT: toward building a foundation model for single cell multi-omics using generative AI

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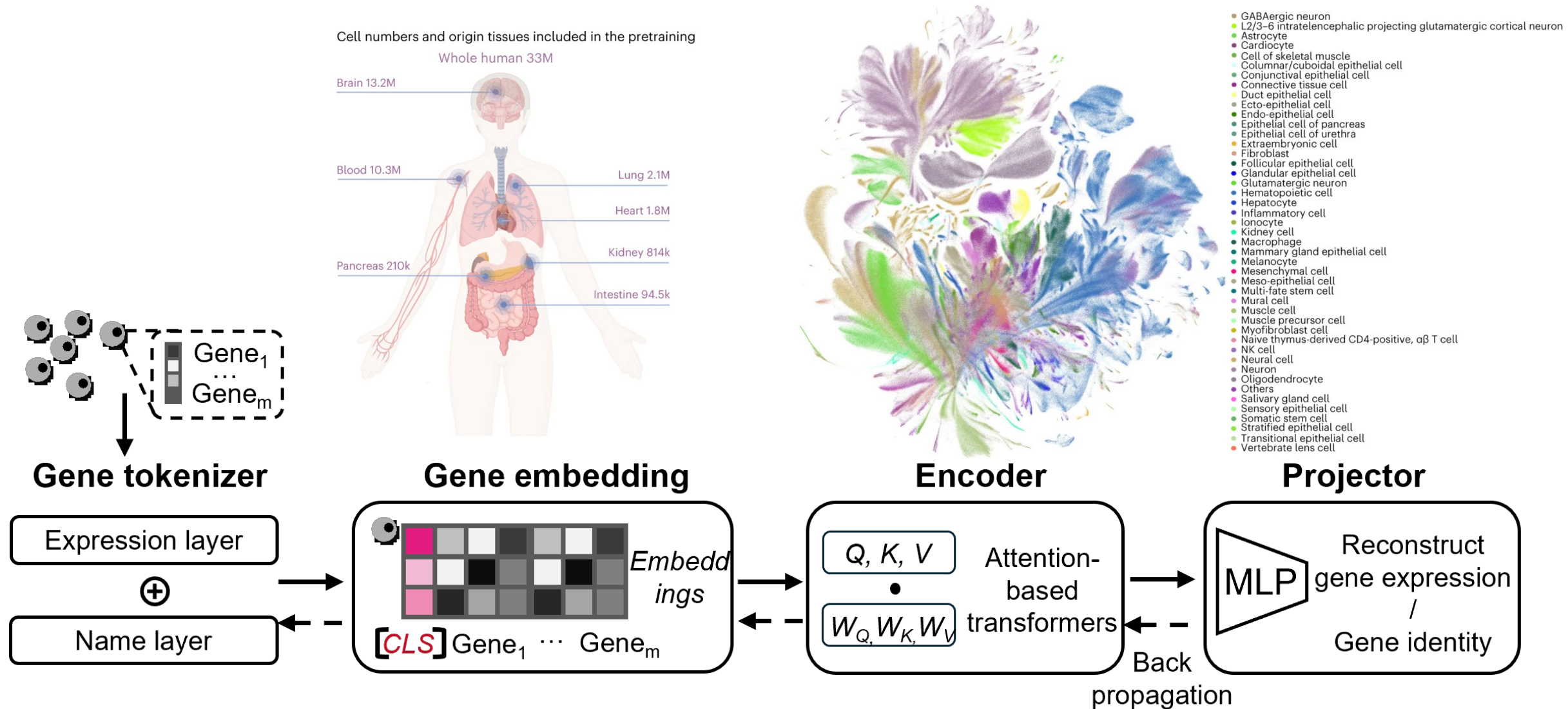
A cell atlas foundation model for scalable search of similar human cells

[Graham Heimberg](#) , [Tony Kuo](#), [Daryle J. DePianto](#), [Omar Salem](#), [Tobias Heigl](#), [Nathaniel Diamant](#), [Gabriele Scalia](#), [Tommaso Biancalani](#), [Shannon J. Turley](#), [Jason R. Rock](#), [Héctor Corrada Bravo](#), [Josh Kaminker](#) , [Jason A. Vander Heiden](#)  & [Aviv Regev](#) 

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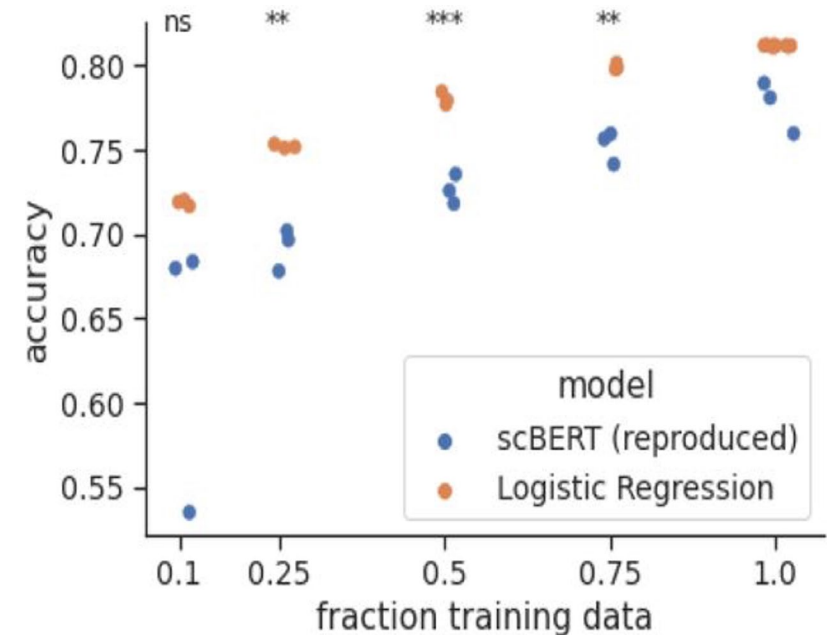
Basic architecture of scLLMs



Slide credit: Cui, H., Wang, C., Maan, H. et al. scGPT: toward building a foundation model for single-cell multi-omics using generative AI. Nat Methods 21, 1470–1480 (2024).

Limited power in out-of-context scenarios

- Do not perform well in zero-shot settings ([Kedzierska, K.Z., et al. *Genome Biol* 26, 101, 2025](#))
 - Excel in in-context scenarios
 - Cannot handle unseen applications
- Need adaptation to train decoders for specific tasks
 - Encoder-only models
- Expensive finetuning might not always outperform simple methods ([Boiarsky, R., et al. *Nat Mach Intell* 6, 1443-1446, 2024](#))
 - Catastrophic forgetting → lose generalizability



Model adaptation: feature distortion theory

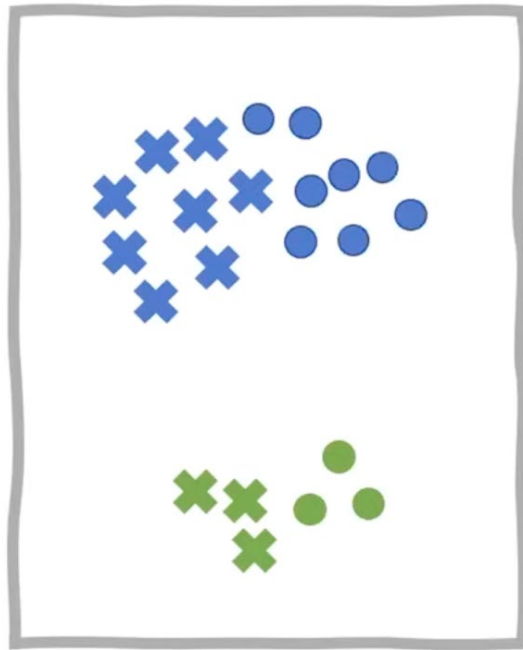
Pretrained
Features

Fine-tuning

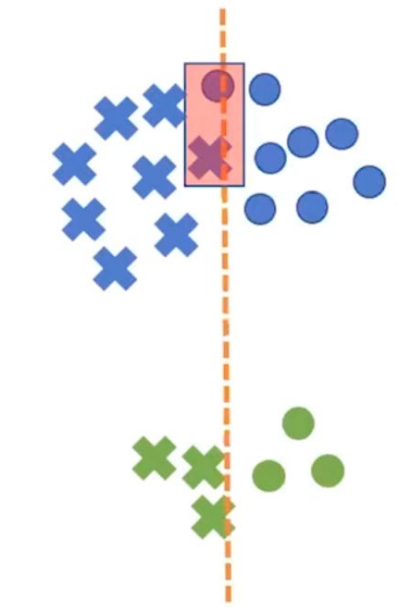
Linear probing: freezes
pretrained features

Finetuning
data

Testing data



Head performs
poorly on OOD
examples



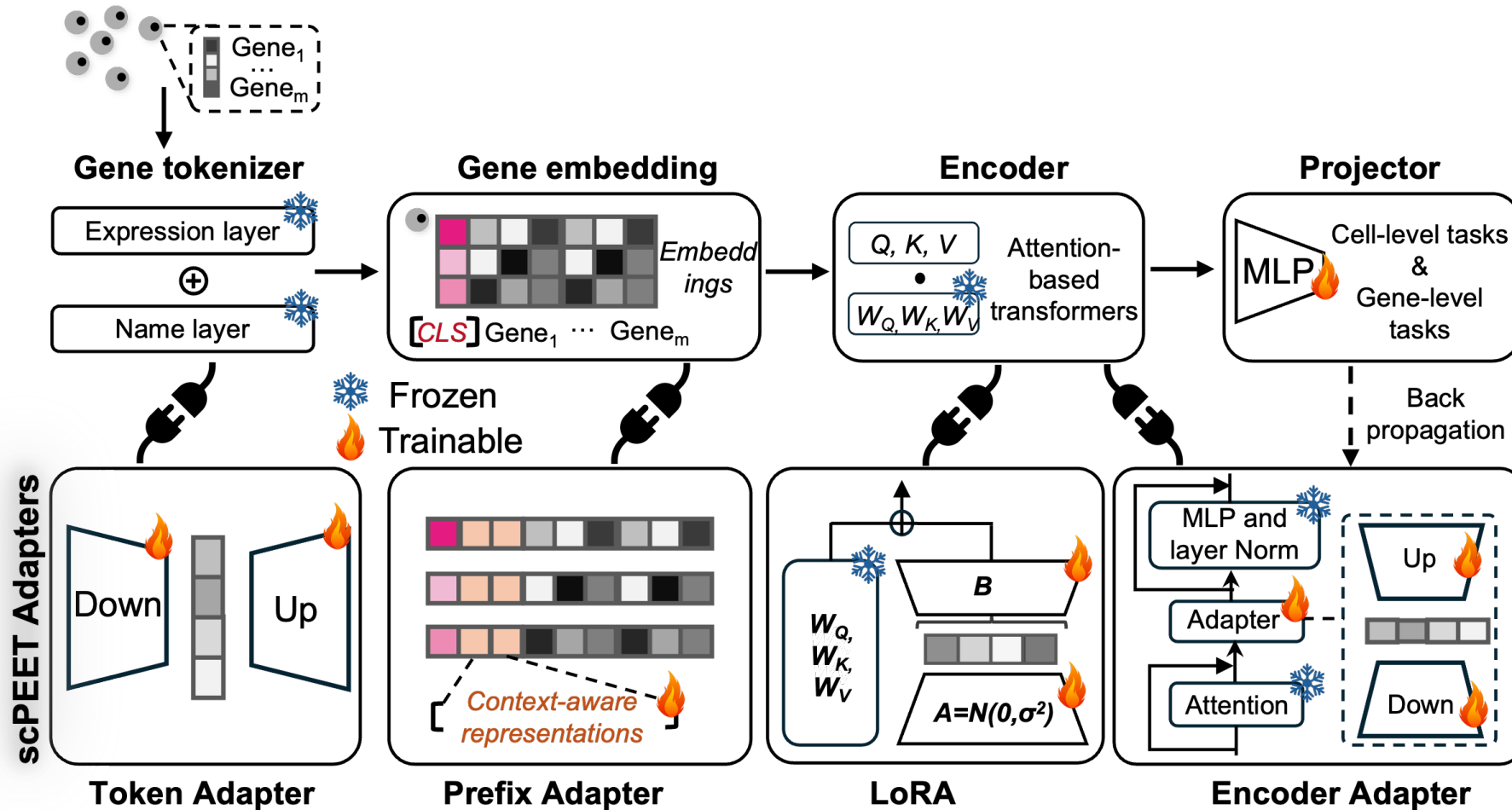
Head is decent on

- Fine-tuning → **catastrophic forgetting** of previously acquired knowledge → lose generalizability

Kumar et al., 2024: <https://openreview.net/pdf?id=UYneFzXSJWh>

<https://snorkel.ai/blog/boost-foundation-model-results-with-linear-probing-fine-tuning/>

Parameter-Efficient Fine-Tuning scLLMs (scPEFT)



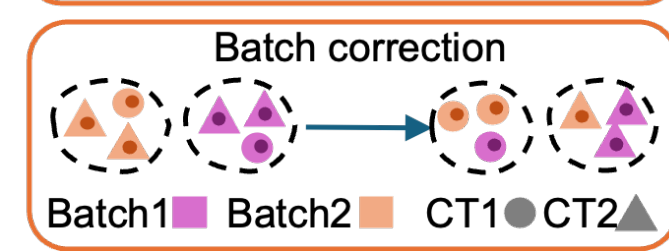
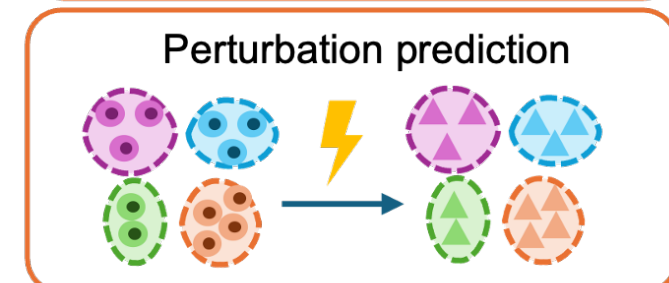
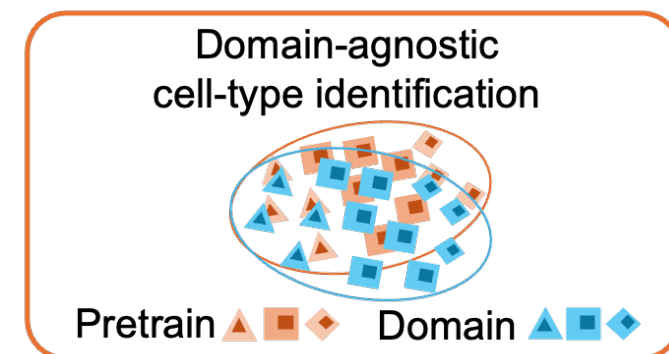
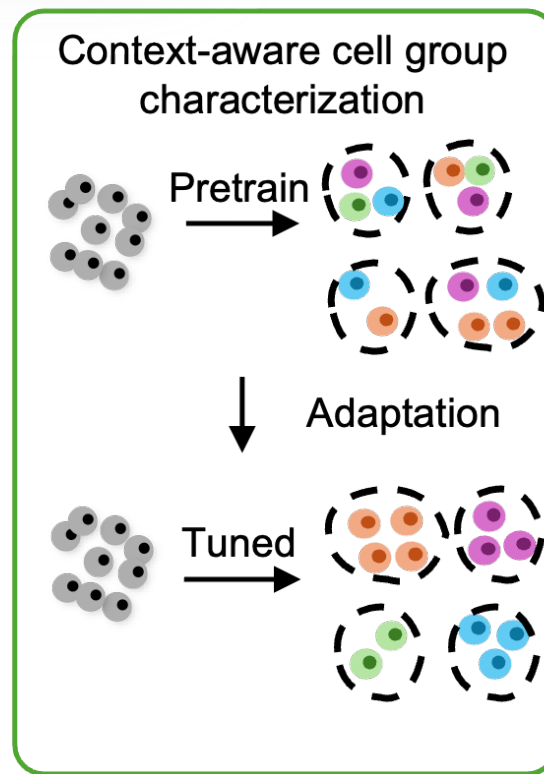
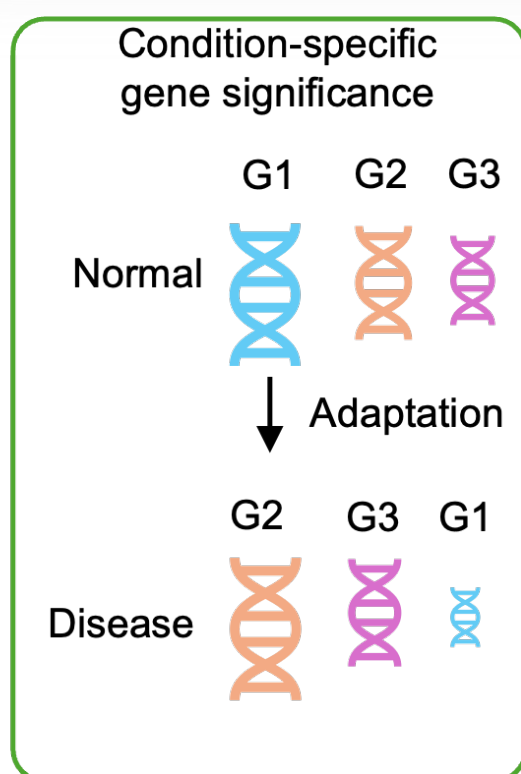
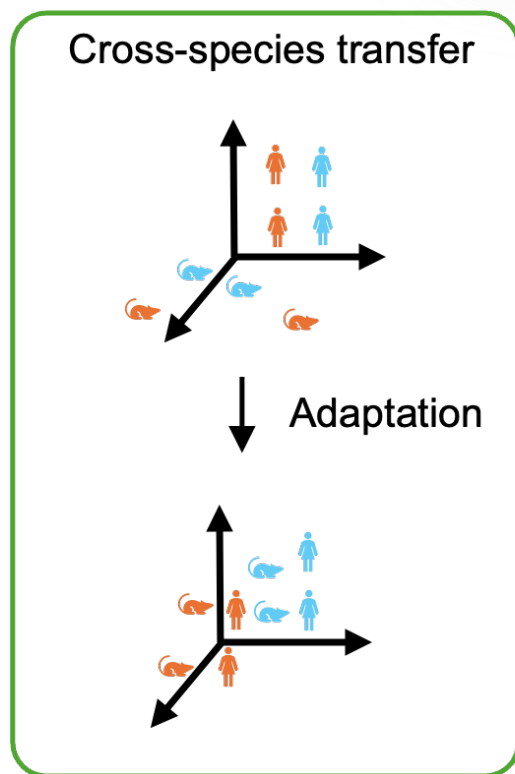
✓ Prevent catastrophic forgetting

✓ Minimize computational demands

He, F., Fei, R., Krull, J.E. *et al.* Harnessing the power of single-cell large language models with parameter-efficient fine-tuning using scPEFT. *Nat Mach Intell* **8**, 118–133 (2026).

Enhance scLLMs in out-of-context applications

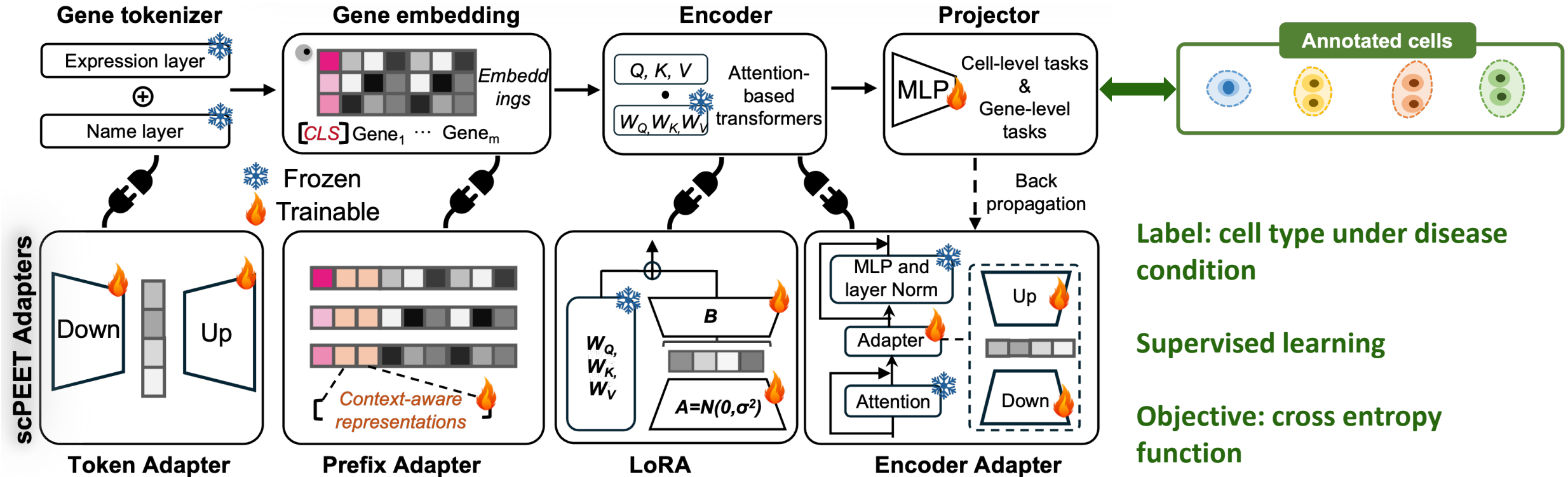
scPEET downstream applications



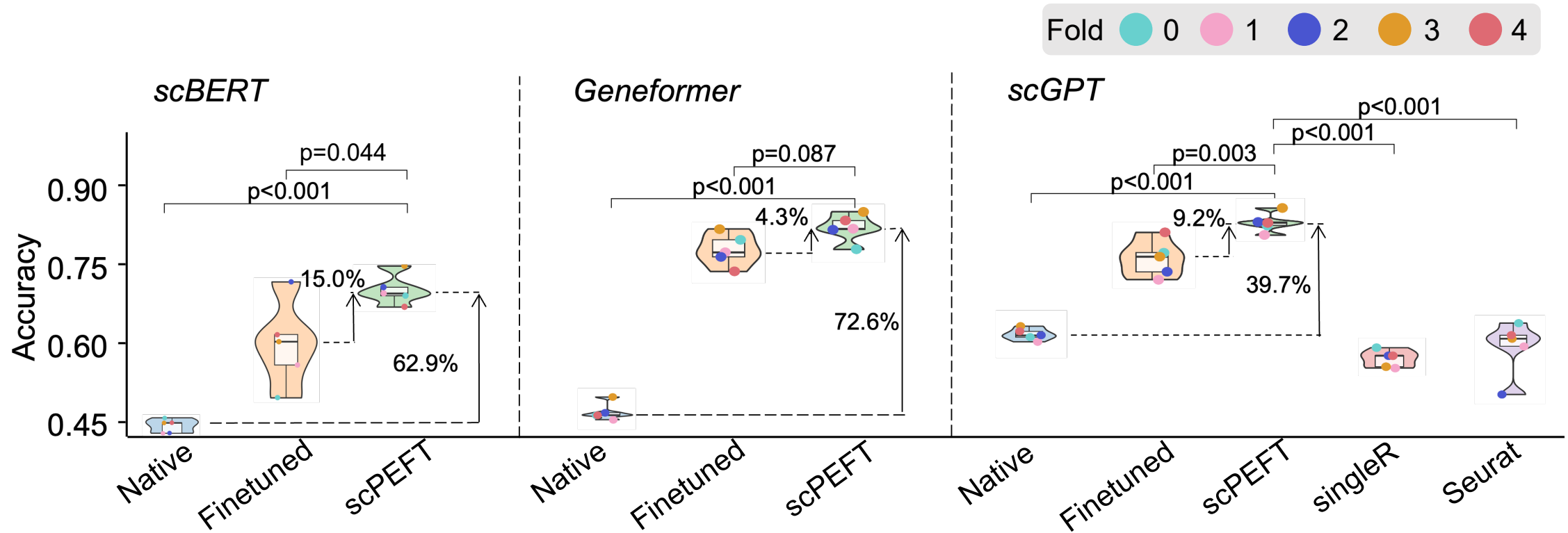
Enable

Enhance

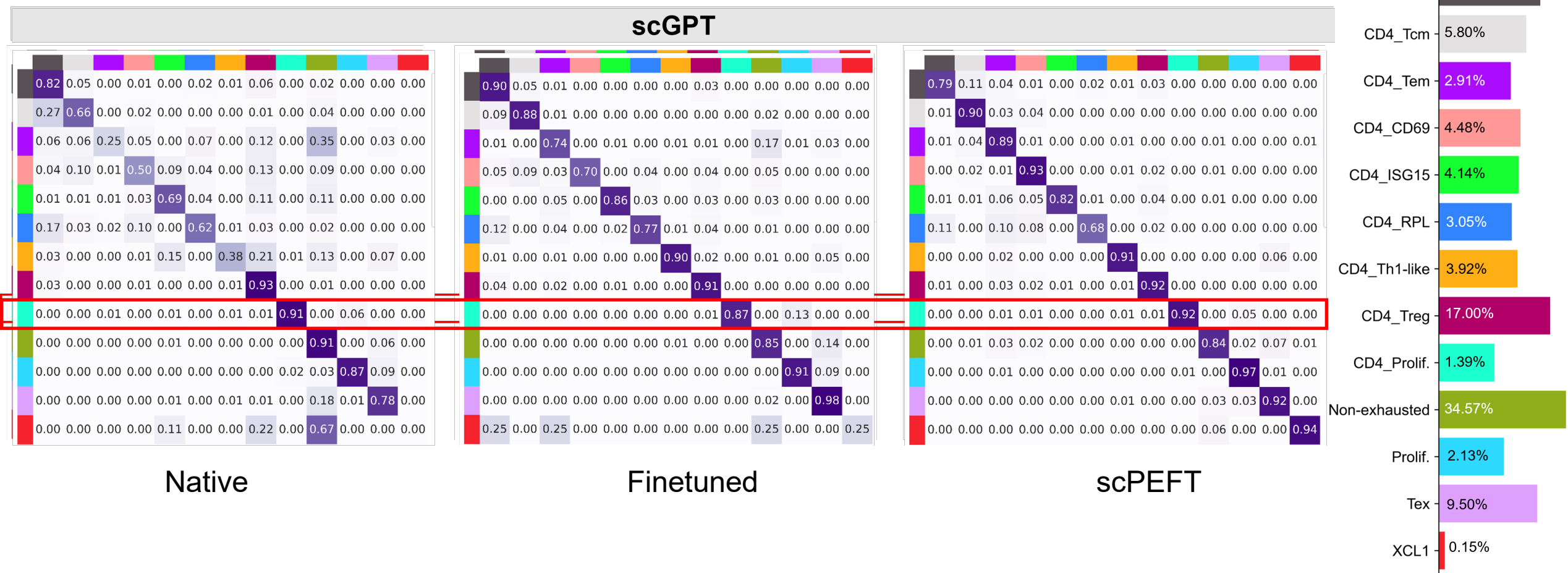
Workflow of cell type identification



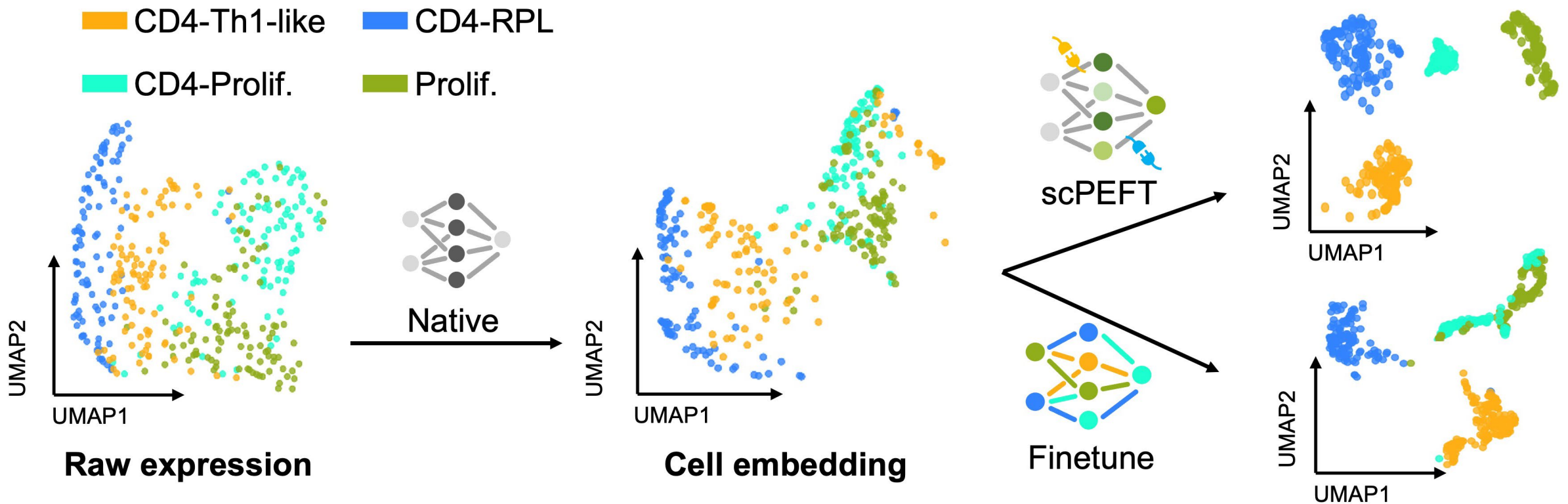
Benchmarking on NSCLC dataset



Confusion matrix on NSCLC dataset



UMAP illustration of catastrophic forgetting



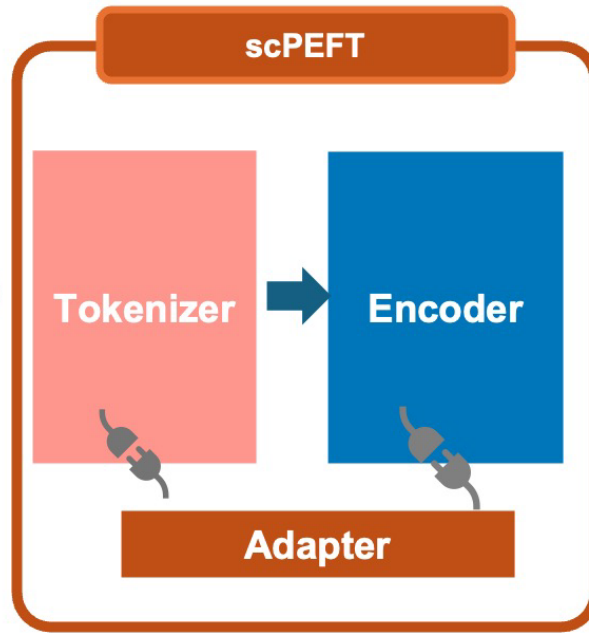
- Randomly extract 10% testing cells from these types

Differential attention analysis of scPEFT

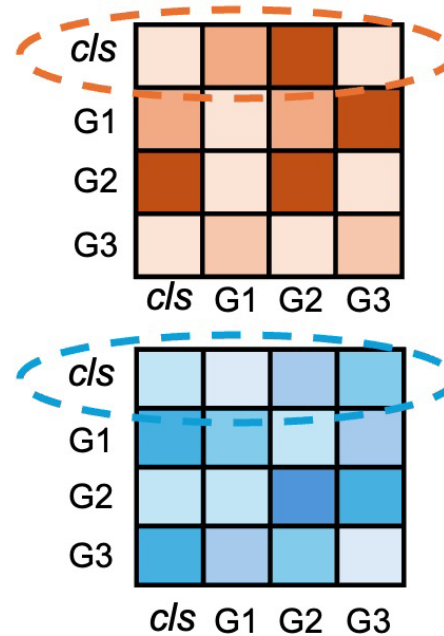
Gene expression of
cell groups
(Target vs. Control)



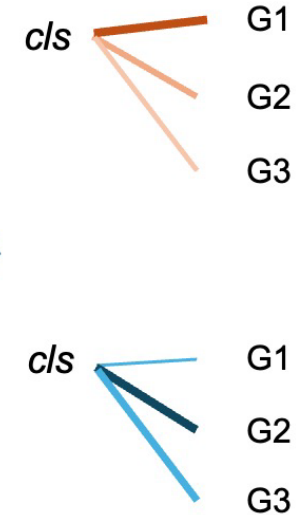
scLLM with tuned
adapters



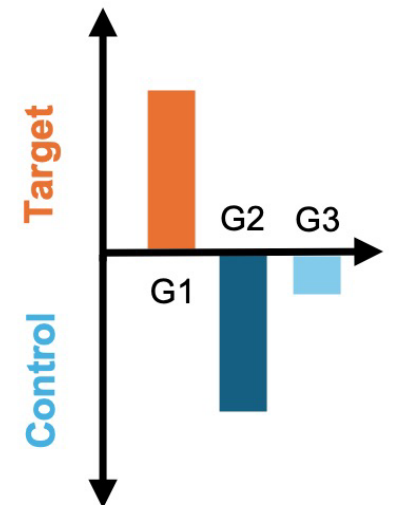
Normalized
attention map



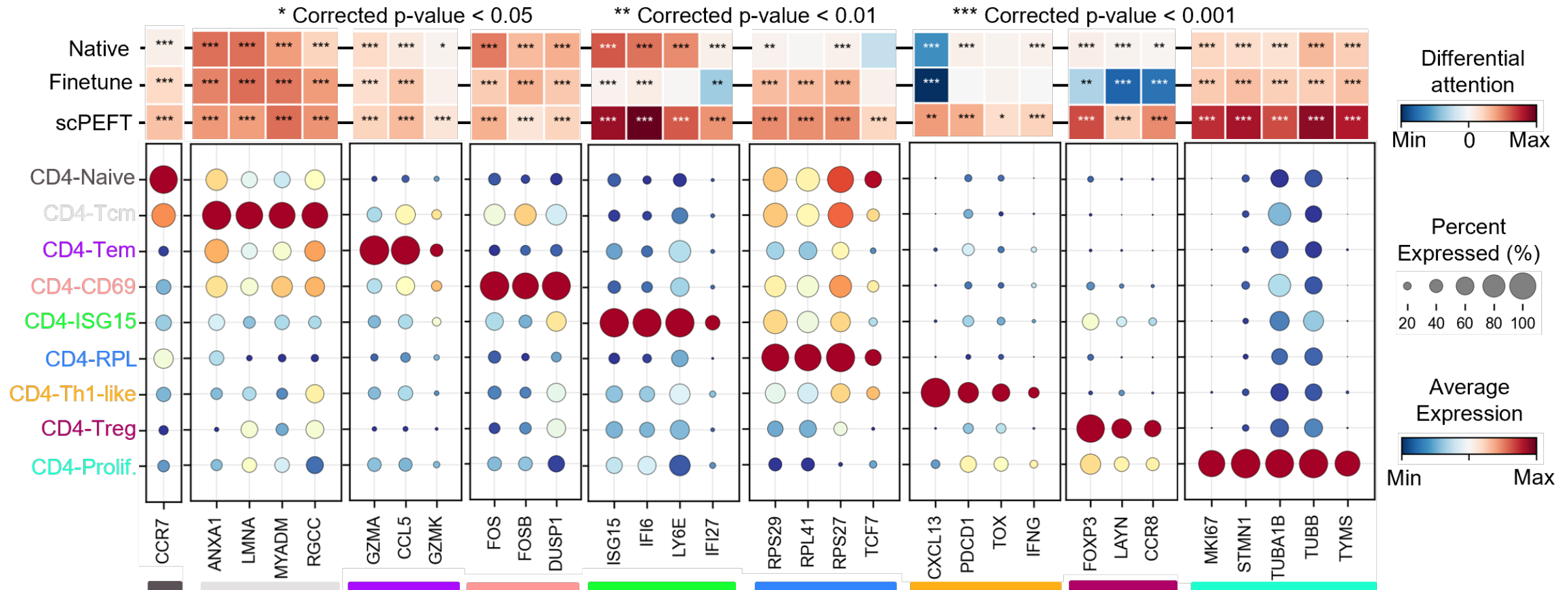
Attentions from *cls*
to gene tokens



Differential
attentions
(Target - Control)



Validation of scPEFT's differential attention scores



Markers source from: Liu, B., Hu, X., Feng, K. et al. Temporal single-cell tracing reveals clonal revival and expansion of precursor exhausted T cells during anti-PD-1 therapy in lung cancer. *Nat Cancer* 3, 2022:108–12.

Condition-specific marker discovery

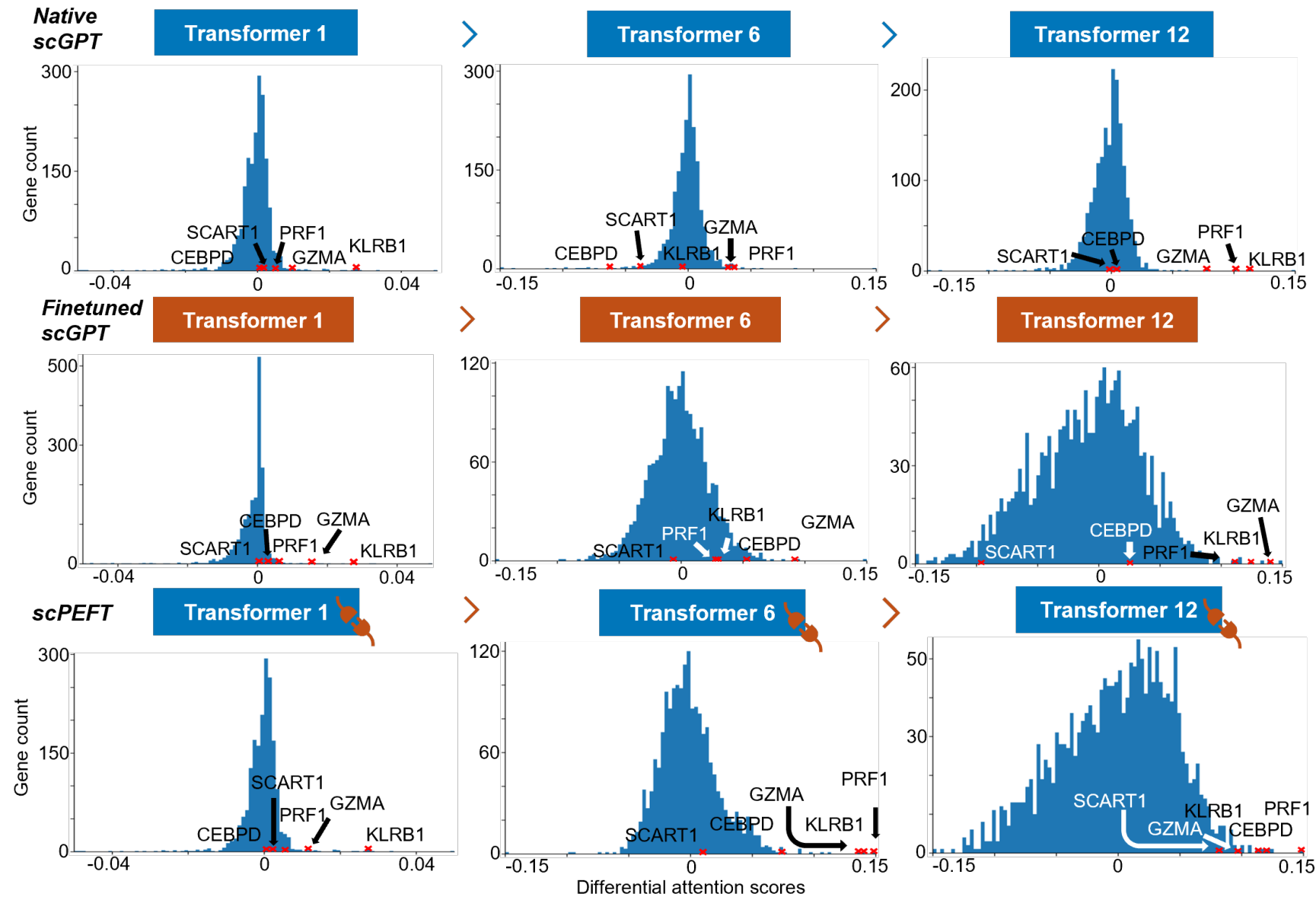
Dataset: COVID

Cell states:

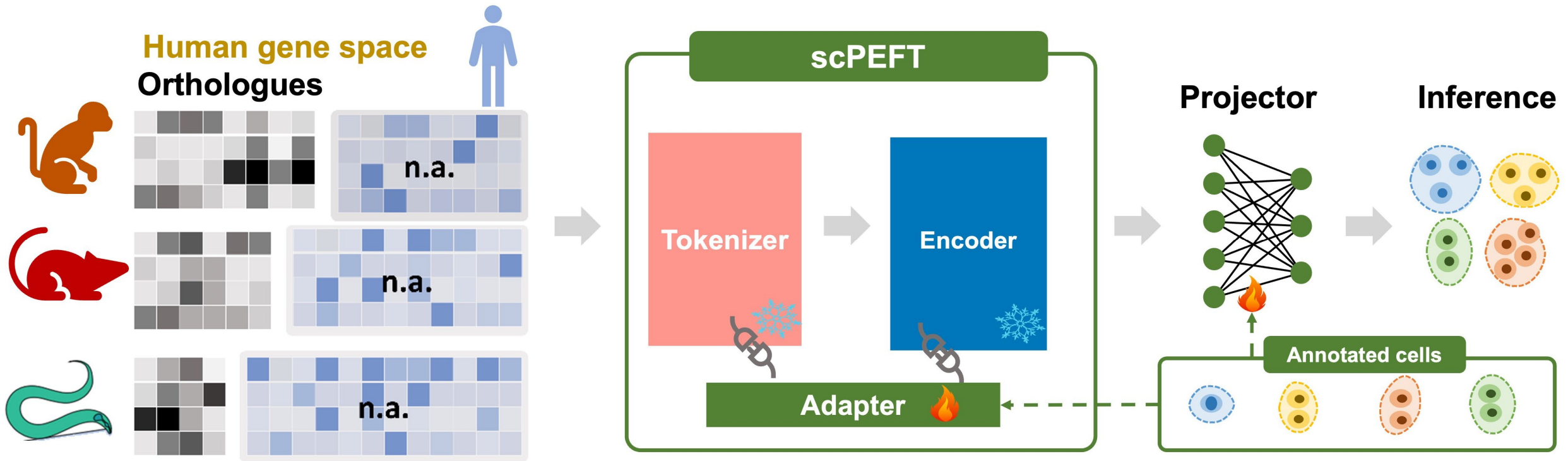
Effector Memory CD8+ T vs
Memory CD8+ T cells

Probes:

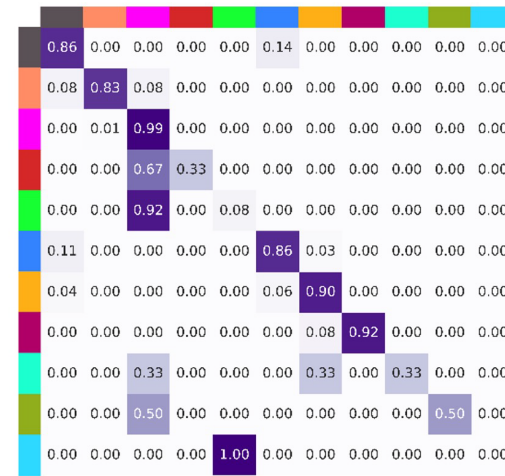
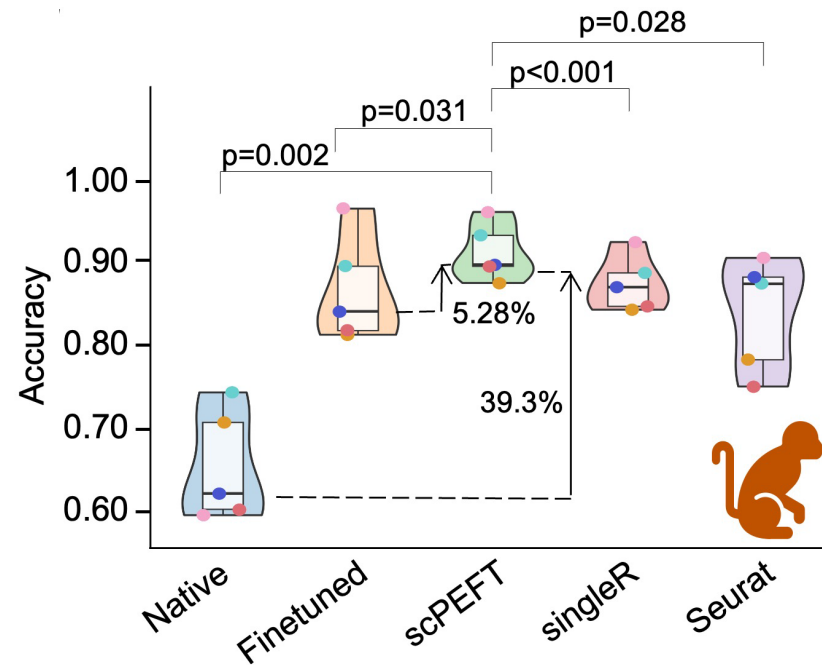
- *KLRB1*, *GZMA*, *PRF1*: Key effector molecules
- *CEBPD* and *SCART1*: Effector-function-associated genes



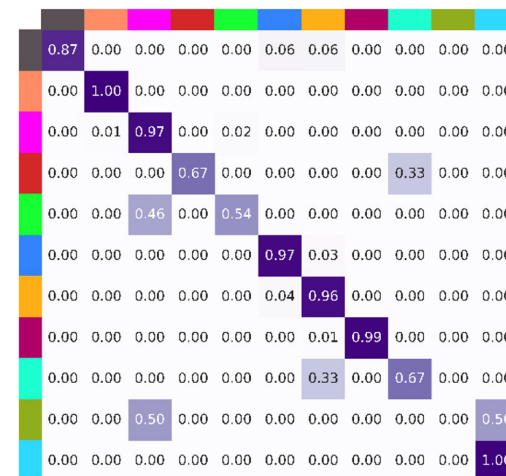
Workflow of cross-species cell type identification



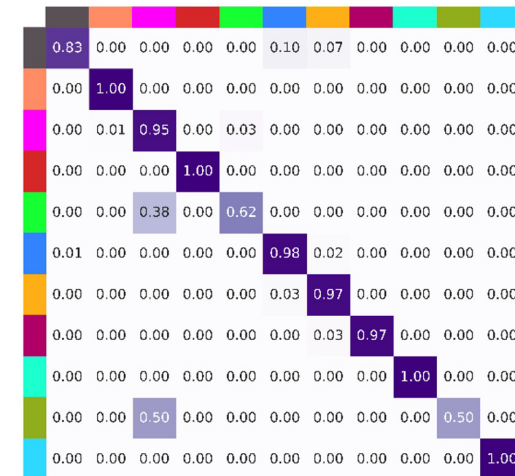
Adaptation of scLLMs to macaque cell identification



Native



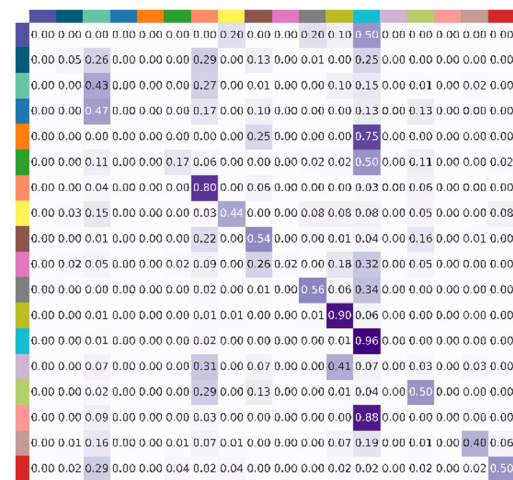
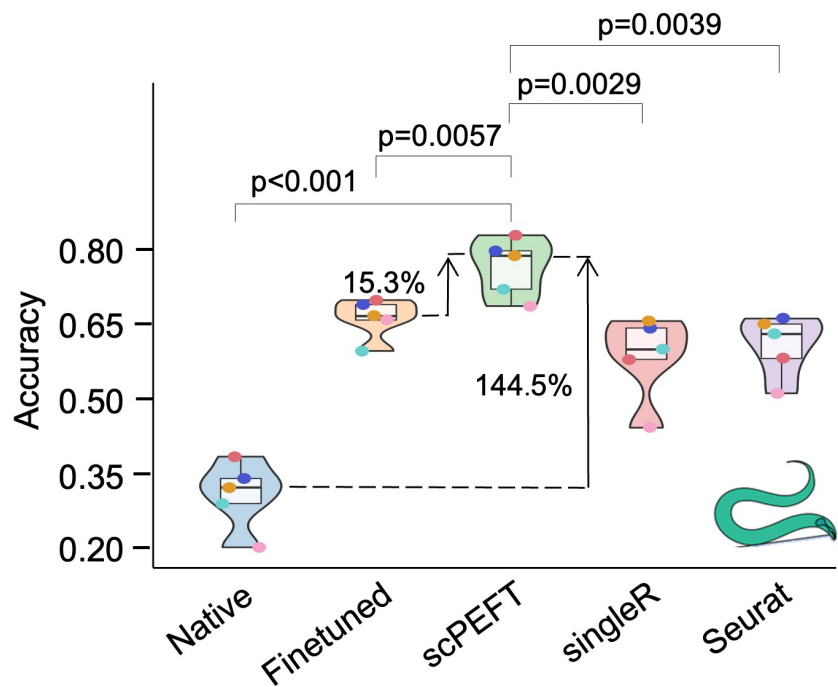
Finetuned



scPEFT

■ Elongating ■ Endothelial ■ ImmLeydig ■ Macrophage ■ Myoid ■ Round Spermatid
 ■ Spermatocyte ■ Spermatogonia ■ Tcell ■ f-Pericyte ■ m-Pericyte

Adaptation of scLLMs to *C. elegans* cell identification

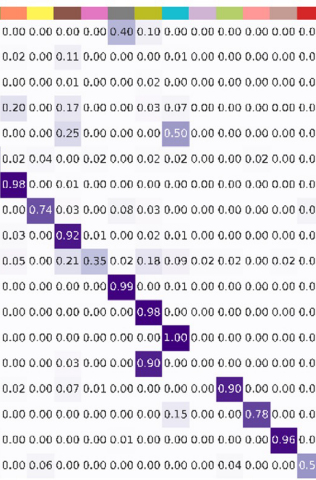


Native

coelomocyte
gonad
seam

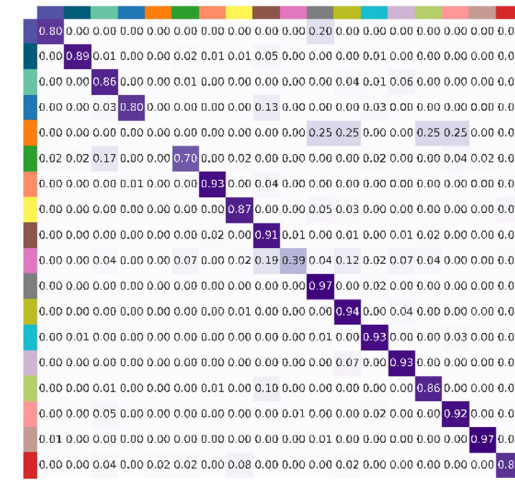
endocrine
hypodermis
valve

expithelium
intestine
various



Finetuned

excretory
mesoderm
vulva



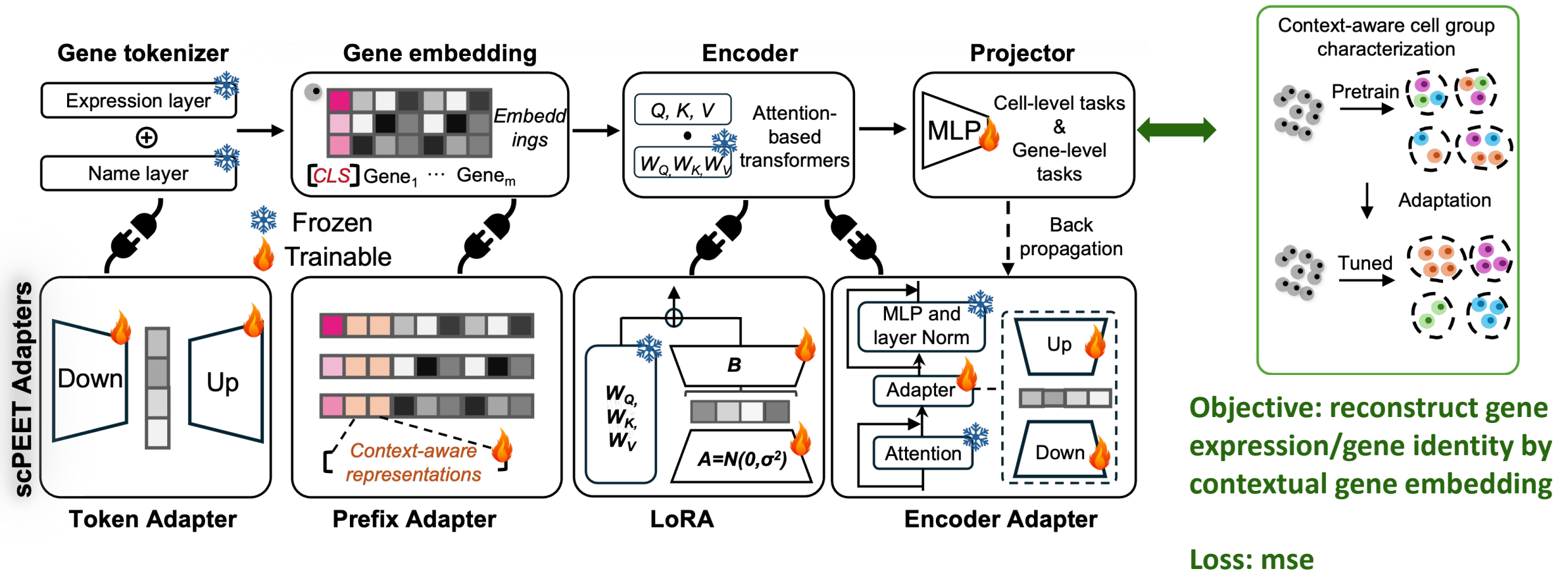
scPEFT

germline
muscle
neuron
pharynx

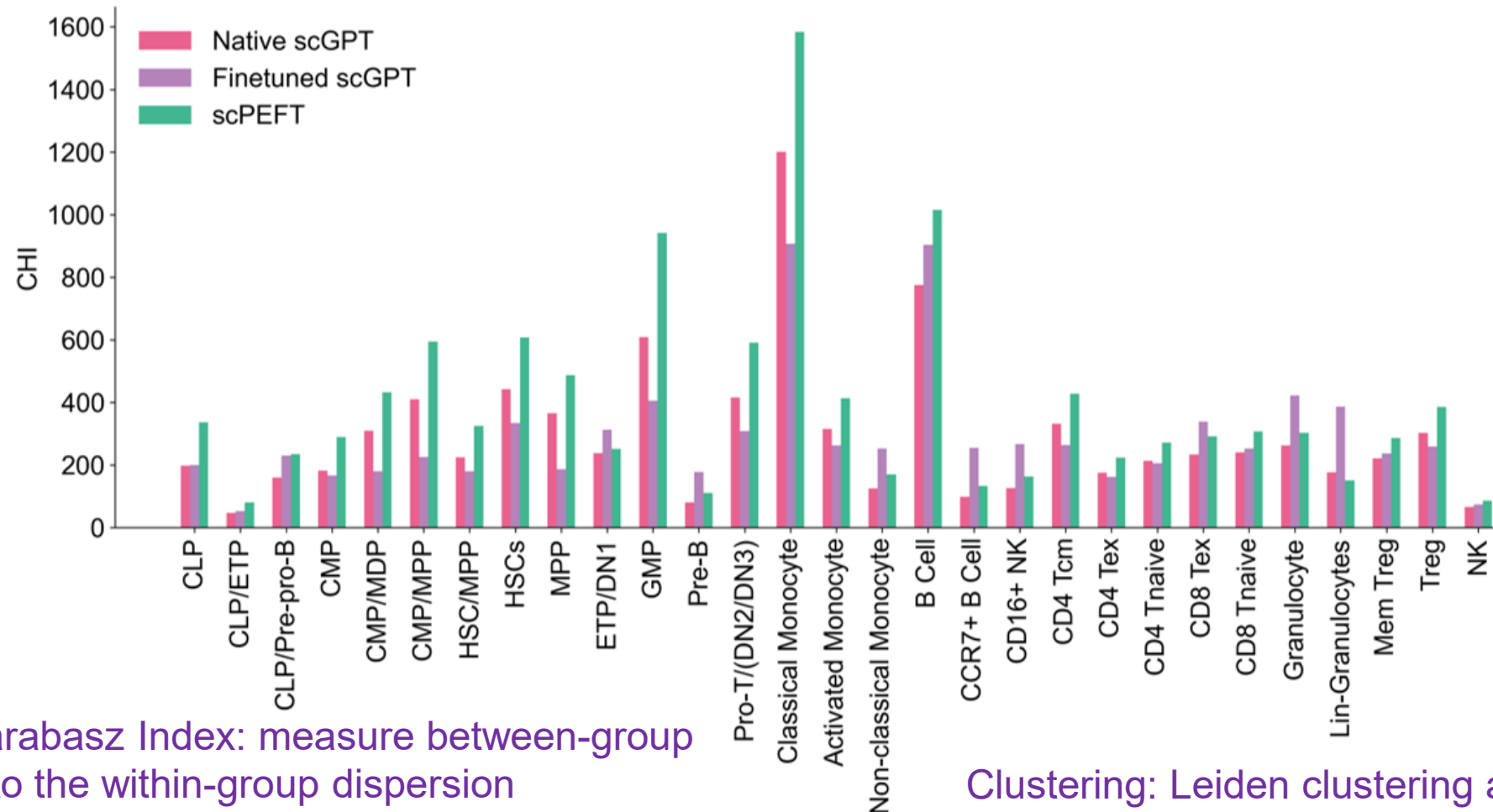
gland
various

glia
vulva

Adaption of scPEFT in unsupervised manner



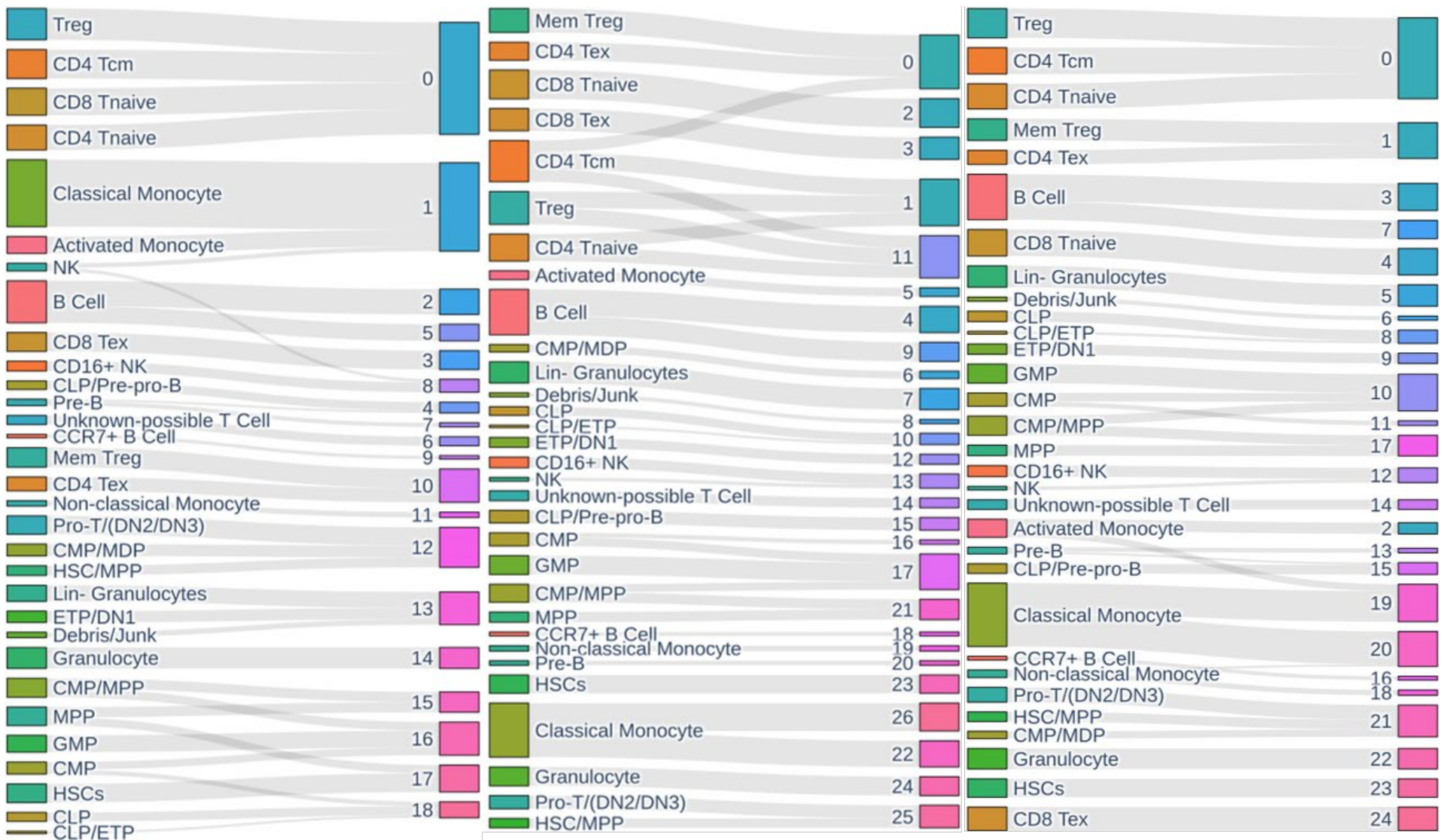
Cell clustering from adapted models

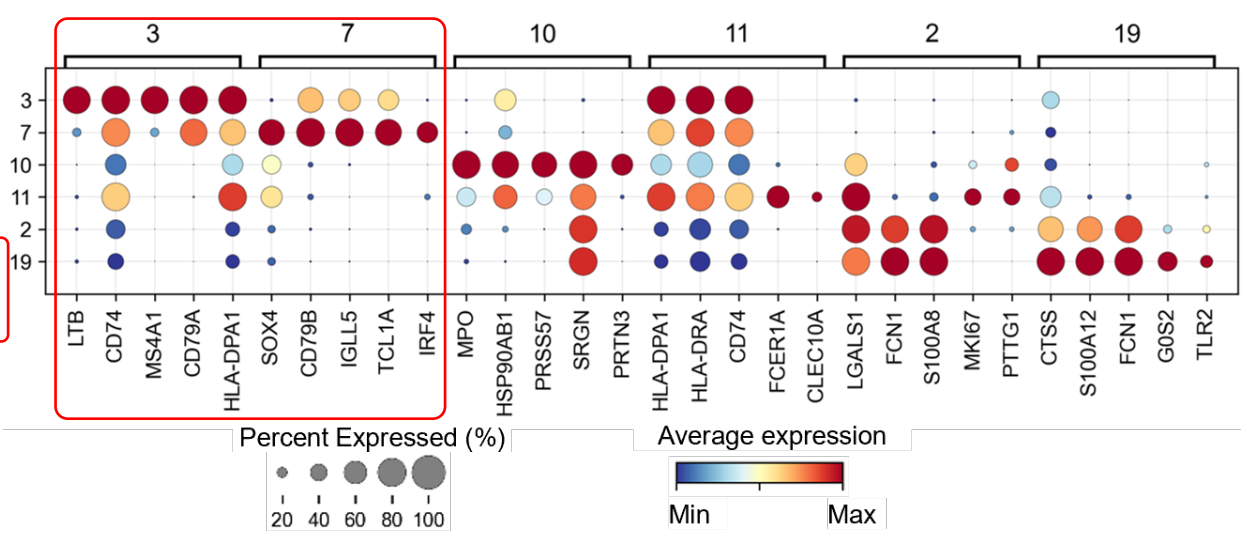
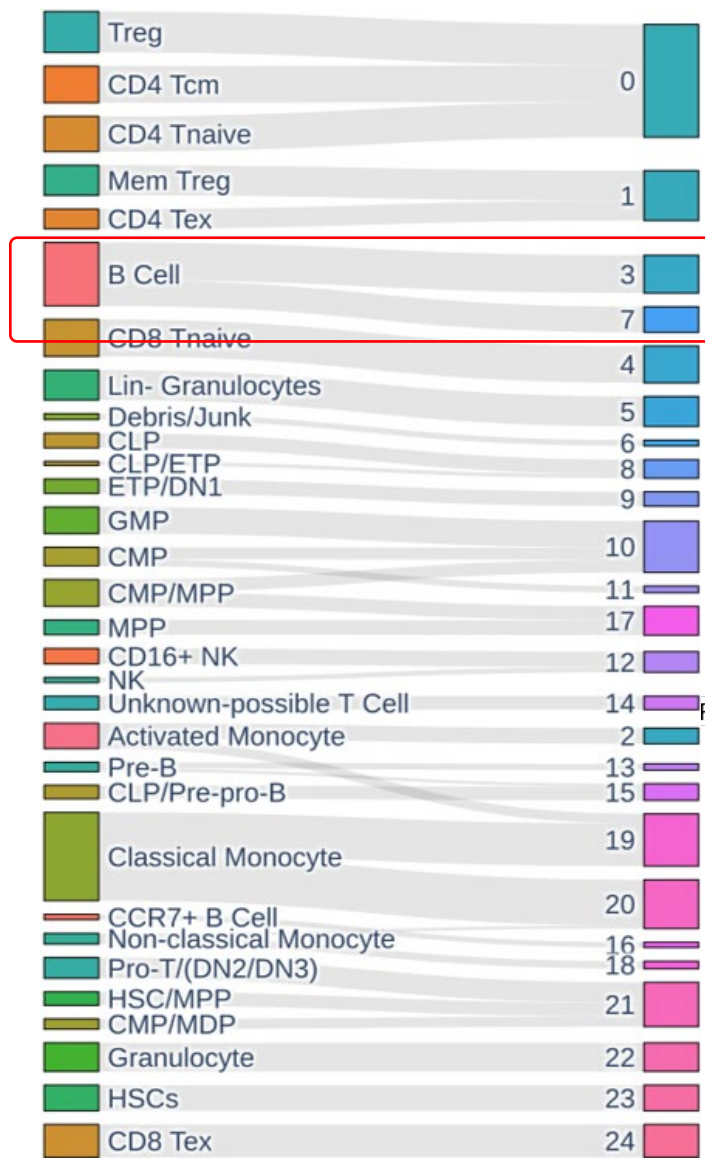


Calinski-Harabasz Index: measure between-group separation to the within-group dispersion

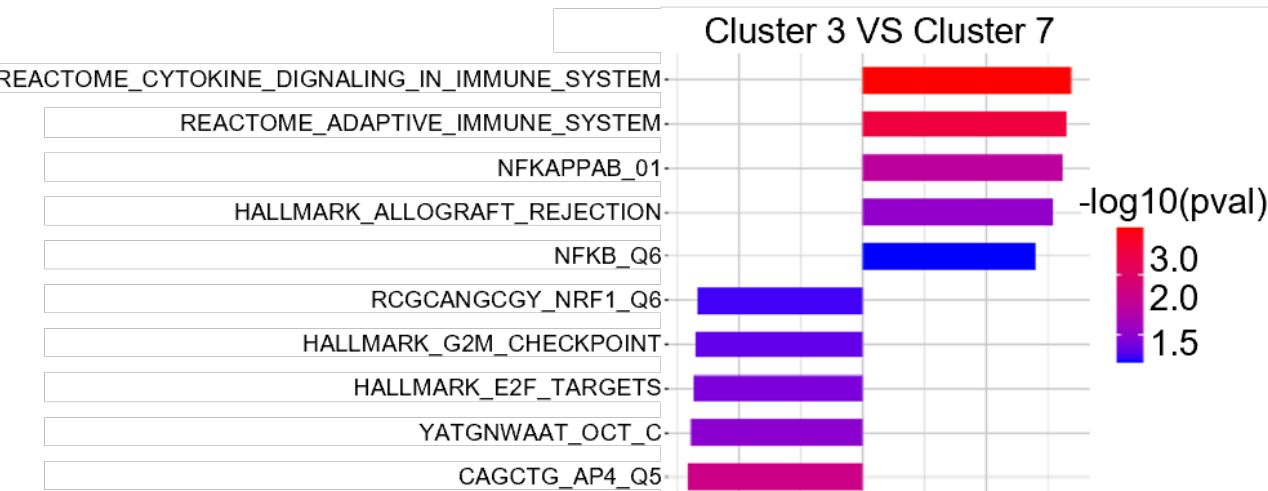
Clustering: Leiden clustering at a resolution of 1.5

Cell groups annotated by surface protein expression





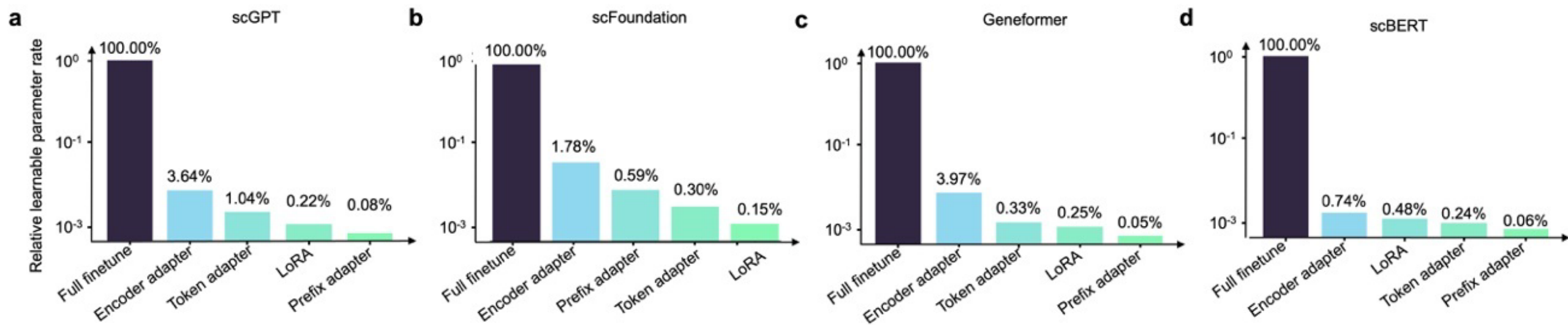
Cluster 3 enriched for *NFkB* target genes and HLA molecules: a recently **developed B cells**



Cluster 7 showed enrichment for pro-growth and developmental pathways (*E2F* and *AP4* targets, *SOX4*, *IRF4*): **activated or memory B cells**

Gene Set Enrichment Analysis with attention scores

Efficiency analysis in parameter usage



- Reduced up to 96% learnable parameters
- Enabled adaptation of scLLMs on single average GPU (Nvidia RTX 3090 with 24GB GPU memory)
- Achieved faster converge and needed less labeled data

Thanks!